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T&M

“Development And Functionality Of INNOVATIONS Will Be SPURRED By SUPERIOR TESTING”

Make in India initiative will lead manufacturers from across the globe to set up shop in India. This will need new test solutions. Andreas Pauly, executive vice president - test and measurement, Rohde & Schwarz GmbH & Co. KG, and Yatish Mohan, managing director, Rohde & Schwarz India, share insights on the latest trends in the test and measurement (T&M) sector, in a conversation with Deepshikha Shukla of *Electronics For You*

Q. Which technology verticals will drive the growth of the T&M market?

A. Increased electronic device adoption will drive demand for T&M, as advanced devices need advanced T&M processes. Development and functionality of innovations will be spurred by superior testing.

Autonomous driving, 5G, the IoT, data centres, power applications and high-speed digital standards are the growth drivers over the next five years.

According to a recent study, till 2022, growth wave will span across technologies and cover entire product life cycles.

Great opportunities will arise from high-frequency applications. These will fundamentally change the face of the T&M market over the next decade.

The industry will also see opportunities in sectors like automotive and industrial. Demand from the medical device sector will go up as well.

In the next 10 years, wireless technology and its applications will see many changes. The IoT is a key driving factor. Emergence of 5G and need for interoperability will change the tech landscape.

Q. How will T&M requirements evolve with the launch of 5G services?

A. 5G New Radio (5G NR) promises gigabit connection speeds, lower latency, enhanced reliability, increased connection density and improved energy efficiency. Key features in 5G NR important for realisation of the same include use of millimetre-wave frequencies (up to 100GHz) and support for significantly

higher signal bandwidths.

Designing power amplifiers that exhibit the necessary characteristics may need new approaches.

Circuitry needed to generate frequencies in millimetre-wave range requires careful design and selection of components to reduce the effects of phase noise introduced by mixers and multipliers.

Beamforming also requires good amplitude and phase synchronisation between antenna elements in active antenna systems.

High frequencies intensify the challenges for T&M. Short wavelengths and higher losses in circuits need tight integration, making it impractical to supply connector ports for testing.

At the same time, effects of connectors and test fixtures become non-negligible, potentially affecting the validity of conducted measurements. As a result, over-the-air testing will play a key role in the 5G space.

Q. How do you see the Indian T&M market?

A. It is transforming. Initiatives like Make in India have opened new revenue streams for T&M in mobile and aerospace and defence manufacturing.

On telecom side, the initial focus was on a few base station manufacturers. This will expand with device manufacturing in the country.

The government has mandated RF, specific absorption rate and EMC testing from Indian test labs for all telecom and IT-related products. Telecom Engineering Centres and associated

industries are gearing up to set up conformance testing labs for devices and network equipment. This makes us hopeful from the test house business perspective.

With Make in India, the industrial components and research market will also grow in India.

Q. How do you describe the presence of Rohde & Schwarz in India?

A. In India, we partner with agencies, training institutes and technology companies to facilitate technology growth in the country. For several years, we have been associated with Indian PSUs for tech transfer in the fields of broadcast transmitters, communication and monitoring receivers. Recently, we collaborated with a defence PSU for manufacturing of radios for air defence applications.

We achieved about 15 per cent growth last year here. We were named Indian T&M Company of the Year at Frost & Sullivan's 2018 India Best Practices Awards Banquet in Mumbai.

Q. How do you cater to academia and research sectors in India?

A. Education is an important segment for us. We have invested in R&D, with a dedicated team of 70 engineers, and have localised our solutions for this sector. We offer cost-effective T&M solutions with specific ranges and features for Indian educational institutes.

We are also thinking of sponsoring reputed institutes by equipping their labs with our test instruments. **EFY**